FM vs AFM order

At $t_1 \ll U_0$ → $H_{\text{eff}} = J \sum_{\langle i,j \rangle} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$ simplest coupling $J \approx 4\frac{t_1^2}{U_0} - 2X$

direct exchange

superexchange

QML grows sublinearly below $(\theta^2/V)^*$



